

Michael L. Green

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Education

Central Michigan University, Mount Pleasant MI
Bachelor of Science in Computer Science, Minor in Mathematics

Graduation: August 2023

Experience

Software Developer Intern

January 2022 - August 2022

24G, Troy MI

- Developed interactive expo activations for Fortune 500 clients including Salesforce, Waymo, and Essilor Luxottica.
- Created the Oakley expo activation that Essilor Luxottica named their favorite of 4 experiences at the event, delivered as the first client assignment of the internship.
- Independently took 4 live experiential activations from UI mockup to deployed production within 2-3 weeks each.
- Built a full-featured analytics dashboard UI for an internal expo reporting tool, including data display and CSV export, that remained in active company use following the internship.
- Engineered touch-aware Vue.js interfaces for experiential activations, engineering for non-standard screen dimensions and touch input patterns across varied expo display hardware.

Projects

Tally Tracker - [Demo Video](#) | [Repository](#)

January 2026 - March 2026

Technologies Used - Vue, Typescript, Ionic, Capacitor, Node, Express, Socket.IO, Prisma, Zod, Cypress, Vitest

- Engineered a real-time collaboration layer for shared counters with server-authoritative participant resolution and client-side state reconciliation to keep multiple devices consistent during concurrent updates.
- Architected an offline-first architecture with optimistic UI, durable command queue, background sync, and server-side idempotency to safely replay mutations after network failures/app restarts.
- Designed access/refresh token auth with rotation and dual delivery (HTTP-only cookies for web, Bearer tokens for native) including automatic refresh/retry and concurrent refresh de-duplication in the API client.

Reaction - [Live Site](#) | [Demo Video](#) | [Client Repository](#) | [Server Repository](#)

September 2025 - December 2025

Technologies Used - Vue, Pinia, Sass, GSAP, Node, Express, Sequelize

- Architected a reactive Vue 3 frontend utilizing Pinia for modular state management and GSAP for high-fidelity timeline animations, significantly enhancing user experience through intuitive visual feedback.
- Developed backend services adhering to the Repository Pattern, creating a clean separation of concerns and improving maintainability.
- Implemented a defense-in-depth security strategy by storing JWTs in HTTP-only cookies to mitigate XSS attacks and applying rate limiting to prevent brute-force attempts.

Game Lobby - [Live Site](#) | [Demo Video](#) | [Repository](#)

June 2025 - July 2025

Technologies Used - Vue, Sass, Pinia, Socket.IO, Node, Express, Prisma, Postgres

- Architected a server-authoritative Node.js backend to centralize game logic, preventing client-side state manipulation and ensuring fair play for multiplayer sessions.
- Orchestrated real-time data synchronization between a Vue.js client and server using Socket.IO, implementing fault-tolerant reconnection strategies via JWTs to preserve user sessions.
- Engineered a real-time lobby system to manage concurrent user connections, enabling dynamic matchmaking and live player status updates.

Technical Skills

Languages - TypeScript, JavaScript (ES6+), SQL, Python, Java, C/C++

Engineering - Real-Time Systems, Offline-First Architecture, User-Centered Engineering, Authentication & Authorization, REST API Design, Automated Testing & Verification, CI/CD, AI-Assisted Development

Technologies - Vue, React, Ionic, Capacitor, Node.js, Express, Socket.IO, PostgreSQL, Prisma, Docker